

Exercise guide: CA

The goal of this activity is to familiarize everyone with CA simulations, using **NetLogo** (<http://ccl.northwestern.edu/netlogo/download.shtml>). There are many models already included in the NetLogo installation. For additional information, see for instance the following link: NetLogo Library of models: <http://modelingcommons.org/>

- 1) Use NetLogo to simulate two chosen 1D Cellular Automata (CA), in two different Wolfram classes. Explain initial conditions and CA rules and evolution. Plot and comment your results.
- 2) Run two different 2D or 3D CA models in NetLogo. Pick models from the library in your own installation (directory *Sample Models*). or from *Modelling Commons*, which runs most models on their web server. For the server, note that NetLogo is not backwards compatible, and you should pick relatively recent models.
 - a) Report CA rules for the chosen models.
 - b) Before running the model for the pre-set conditions: what kind of results do you expect?
 - c) Enlarge your “universe”. Note that the default boundary conditions in NetLogo are periodic boundary conditions. Is the change leading to new behavior?
 - d) Most models require setting a random initial condition. Start with different random instances. Do you notice different qualitative results? Explain.
 - e) NetLogo uses its own programming language. Check the code for the examples you picked. For one of the chosen models, change the code, for instance modifying one of the rules and report what happens when you run with that new set-up.